

Architecting Disaster Recovery & High Availability on AWS

EC2 AMI Recovery · EBS Backup & Restore · RDS Multi-AZ · DB Snapshots · DR Strategies

RTO

Minutes

Fast Recovery

RPO

Minimal

Data Loss

SLA

99.9%

Availability

Layers

3-Tier

DR Coverage

Production-Grade · Portfolio-Ready · Interview-Focused

What We Cover Today

01 Real-World Problem Statement

Why DR is non-negotiable in production

02 Architecture Overview

3-layer DR: Compute, Storage, Database

03 Core Concepts

RTO, RPO, HA, Fault Tolerance, Failover

04 DR Strategies

Backup & Restore → Active-Active

05 Project 1 — EBS DR

Volume backup, snapshot, restore

06 Project 2 — EC2 + RDS DR

AMI recovery + Multi-AZ + DB Snapshots

07 Interview Q&A

20+ real interview questions answered

08 Key Learnings & Outcomes

Production best practices summary

Real-World Problem Statement

What happens when your production server crashes at 2 AM?

← Downtime impacts revenue, SLA commitments and customer trust.

⚠ Disaster Scenario

- ⚠ EC2 production server terminates unexpectedly
- ⚠ EBS data volume gets corrupted / accidentally deleted
- ⚠ RDS database crashes or becomes unreachable
- ⚠ Entire AWS Region goes offline (regional outage)
- ⚠ Ransomware encrypts production data

Business Impact Without DR

Avg Cost/Hour Downtime	\$300K+
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Data Loss Risk	Permanent
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Customer Trust	Destroyed
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Regulatory Fine	Possible
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✓ The Solution: 3-Layer Disaster Recovery Architecture

✓ Compute Layer (EC2 + AMI)

✓ Storage Layer (EBS + Snapshots)

✓ Database Layer (RDS + Multi-AZ + Snapshots)

Architecture Overview

PRODUCTION-STYLE DISASTER RECOVERY & HIGH AVAILABILITY ARCHITECTURE ON AWS

Backup, Recover, Stay Available.

This architecture demonstrates how we protect, backup and recover three critical layers of an application environment on AWS:

Compute (EC2 + AMI), Storage (EBS + Snapshots) and Database (RDS + Multi-AZ + Snapshots).

Designed For
Business Continuity, DR Readiness
and High Availability

RTO (Recovery Time Objective)
Low Recovery Time
(Fast Recovery using AMI & Snapshots)

RPO (Recovery Point Objective)
Minimal Data Loss
(Point-in-Time Recovery Capability)

Availability Goal
R5P999 (99.999% Availability)

WHAT THIS ARCHITECTURE COVERS

COMPUTE LAYER (EC2 + AMI)

- EC2 instances for application server
- AMI captures for backup
- EC2 recovery from AMI

STORAGE LAYER (EBS + Snapshots)

- EBS volumes for persistent data
- Snapshots capture (Point-in-Time)
- Volume restoration from snapshot

DATABASE LAYER (RDS + Multi-AZ)

- Amazon RDS (MySQL) with Multi-AZ
- Automated standby for failover
- DB Snapshots for backup
- DB restore from snapshot

KEY BENEFITS

- High Availability with Multi-AZ RDS
- Data Availability with Snapshots
- Fast Recovery using AMI & Snapshots
- Cost-Effective Backup Strategy
- Built-in AWS Services (No Third Party)
- Scalable and Elasticity-Ready

AWS

Primary Region

COMPUTE LAYER

EC2 Instance
(Production Server)

AM (Cloud-Init)

EC2 Instance
(Recreated from AMI)

(In case of failure)

STORAGE LAYER

EBS Volume
(Attached to EC2)

EBS Snapshot
(Point-in-Time Backup)

EBS Volume
(Restored from Snapshot)

DATABASE LAYER

Amazon RDS MySQL
(Multi-AZ)

Multi-AZ Standby
(Automatic Failover)

DB Snapshot
(Automated/Manual)

RDS Instance
(Recreated from Snapshot)

BACKUP & RECOVERY

EBS Snapshot
(Stored in Amazon S3)

AMI Backup
(Machine Image Backup)

EBS Snapshot
(Automated/Manual Backup)

DISASTER RECOVERY WORKFLOW

1. Disaster Occurs
(Hardware failure, data loss, or corruption)

2. Identify Recovery Point
(Determine state of system)

3. Restore Resources
(Launch EC2 from AMI, Restore EBS Volume, Restore RDS from Snapshot)

4. Validate & Test
(Verify application, RTO and RPO compliance)

5. Resume Operations
(Application back online)

NEW BACKUP LAYER: ARCHIVES ON S3

Amazon S3 with AWS

- Create AMI/EC2 instances
- For cases of failure, launch new instance from AMI
- Quick recovery of system (disaster)

Storage (S3 with Snapshots)

- EBS Snapshot captures exact volume state
- Restore volume from snapshot
- Reboot/launch EC2 and connect DB

Database (RDS with Multi-AZ)

- Multi-AZ provides automatic failover to standby instance
- Snapshot provides point-in-time recovery
- Restore DB when required

EBS Snapshot
Single, consistent snapshot stored in S3

AMI Backup
On-demand AMI creation (Machine Image Backup)

RDS Snapshot
Automated daily snapshots (Stored in S3)

COMPUTE RECOVERY
Launch EC2 from latest AMI

STORAGE RECOVERY
Restore EBS volume from snapshot

DATABASE RECOVERY
Restore RDS instance from snapshot in new Multi-AZ Subnet

NOTES

- All backups are stored in Amazon S3 (managed by AWS)
- Snapshots are incremental and cost-effective
- Multi-AZ provides continuous availability for database
- This architecture is simple, reliable and pay-as-you-go

* This architecture is built using native AWS services and follows best practices for Backup, Disaster Recovery and High Availability.

Legend: Backup Flow, Recovery Flow, Data Flow

COMPUTE LAYER

- EC2 Instance — production application server
- AMI (Amazon Machine Image) — full server snapshot backup
- AMI captures: OS, configs, installed apps, EBS snapshots
- Recovery: Launch new EC2 from AMI in minutes
- DR Region: Copy AMI to another region for regional failover

STORAGE LAYER

- EBS Volume — persistent block storage attached to EC2
- EBS must be in the same AZ as the EC2 instance
- EBS Snapshot — point-in-time backup stored in S3 (managed by AWS)
- Snapshots are incremental — only changed blocks stored
- Restore: Create new volume from snapshot, attach to EC2

DATABASE LAYER

- Amazon RDS MySQL — managed relational database service
- Multi-AZ: Synchronous standby replica in another AZ
- Automatic failover ~1-2 min if primary DB fails
- DB Snapshots — manual or automated point-in-time backups
- Restore: Create new RDS instance from snapshot

Core Concepts Every Cloud Engineer Must Know



RTO — Recovery Time Objective

How FAST must the system recover after a failure?

Example: RTO = 15 min → app must be back online within 15 minutes of failure.

📌 Lower RTO = More infrastructure running in DR



RPO — Recovery Point Objective

How much DATA LOSS is acceptable after a failure?

Example: RPO = 5 min → losing last 5 minutes of data is acceptable.

📌 Lower RPO = More frequent backups / replication

Backup vs High Availability vs Fault Tolerance

Concept	Definition	Downtime?	Example in AWS
Backup	Copy of data stored safely	Yes, recovery needed	EBS Snapshots, RDS Snapshots
High Availability	System continues if component fails	Minimal / seconds	RDS Multi-AZ, ELB
Fault Tolerance	Zero downtime even during failure	None	Multi-Region Active-Active

DR Strategies — From Cheapest to Most Resilient

1

Backup & Restore

Only backups stored (AMIs, snapshots). Launch everything fresh during disaster. Cheapest but slowest.

🕒 **RTO: Hours**

\$ **Cost: \$**

Infra: Nothing pre-running

Use: Non-critical, cost-sensitive workloads

2

Pilot Light

Minimal core infra (usually DB) always running in DR. App servers launched when needed from AMIs.

🕒 **RTO: Minutes–Hours**

\$ **Cost: \$\$**

Infra: Core DB running only

Use: Moderate criticality, startup cost balance

3

Warm Standby

Full but smaller environment always running. Scale up during disaster. Fast recovery, higher cost.

🕒 **RTO: Minutes**

\$ **Cost: \$\$\$**

Infra: Scaled-down full env

Use: Business-critical, e-commerce, banking

4

Active-Active (Multi-Site)

Identical environments running in multiple regions simultaneously. Near-zero downtime. Most expensive.

🕒 **RTO: Seconds**

\$ **Cost: \$\$\$\$**

Infra: Full infra both regions

Use: Mission-critical, financial systems, global SaaS

COST ↑ As DR maturity increases → SPEED ↑ RELIABILITY ↑

EBS Backup & Disaster Recovery

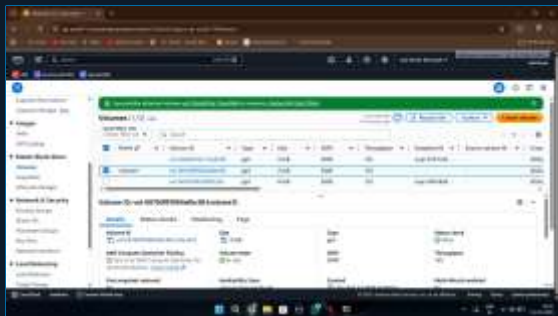


Step-by-Step Implementation

1	Create EBS Volume	<code>EC2 → Volumes → Create volume (2 GiB, gp3)</code>	Separate data disk independent of root
2	Attach to EC2	<code>Actions → Attach volume → /dev/xvdb</code>	OS recognizes it as a new disk
3	Format & Mount	<code>sudo mkfs -t ext4 /dev/xvdb sudo mount /dev/xvdb /data</code>	Make disk usable as a filesystem
4	Write Test Data	<code>echo "EBS backup demo" sudo tee /data/testfile.txt</code>	Simulate production data on disk
5	Create Snapshot	<code>EC2 → Snapshots → Create snapshot</code>	Point-in-time backup stored in S3
6	Restore from Snapshot	<code>Create volume from snapshot → Attach → Mount /restore</code>	Full disaster recovery validation

Project 1 — Screenshots Walkthrough

EBS Volume Created (2 GiB gp3)



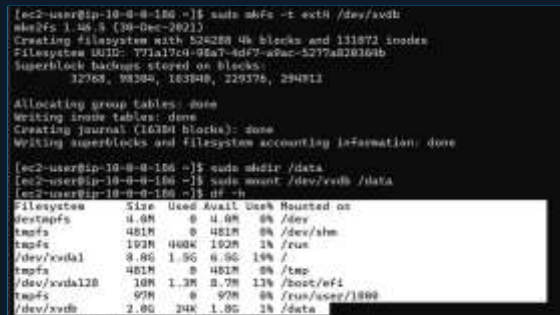
Volume Attached to EC2



Disk Detected via fdisk -l



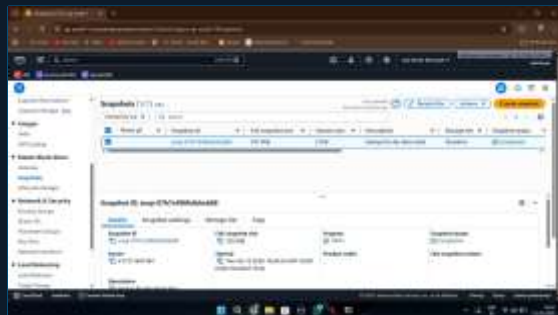
Volume Formatted & Mounted



Test Data Written to /data



Snapshot Created (101 MiB)

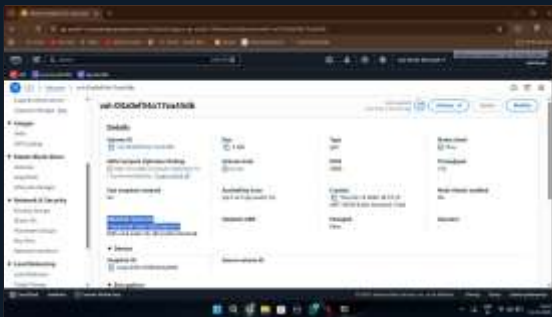


Project 1 — Recovery Screenshots

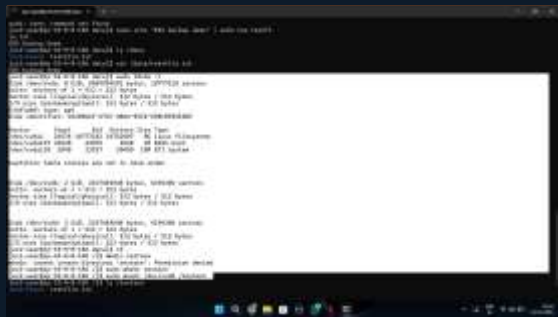
New Volume Created from Snapshot



Restored Volume Attached to EC2



Data Recovered: testfile.txt found



Recovery Outcome

After simulating a disk failure, we successfully restored the EBS volume from snapshot and verified that testfile.txt with content 'EBS backup demo' was intact in /restore. This proves the entire backup-and-restore workflow works end-to-end.

⚠ Important: EBS volumes must be in the SAME Availability Zone as the EC2 instance. EBS snapshots are stored internally in Amazon S3 (managed by AWS — you never see the S3 bucket). Snapshots are INCREMENTAL — only changed blocks are saved after the first snapshot.

EC2 Disaster Recovery Using AMI



What is an AMI?

-  Full server backup including OS + software + config
-  Automatically captures EBS volume snapshots
-  Can be copied across AWS Regions
-  Launch a brand-new EC2 from AMI in minutes

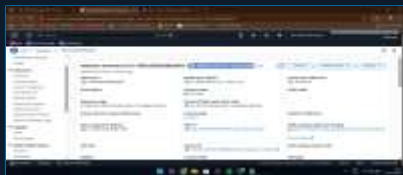
AMI vs EBS Snapshot

AMI	EBS Snapshot
Full EC2 backup	Disk-only backup
Includes OS + apps	Just data/filesystem
Launch new EC2	Attach as new volume
Region-copyable	Region-copyable

EC2 Recovery — Step by Step

Step 1

Launch Production EC2



Deploy DR-demo-server, confirm Running state

Step 2

Create AMI Backup



Actions → Create Image → 'DR-demo-server-backup-ami'

Step 3

AMI Available



Wait for status: Available (AMI includes EBS snapshot)

Step 4

Terminate EC2



Simulate disaster — terminate the production server



Interview One-Liner: An AMI contains instance config + EBS snapshots, but VPC and Security Groups must be re-configured when launching.

RDS High Availability (Multi-AZ) & Snapshot Recovery



RDS Multi-AZ — How It Works

-  AWS creates a SYNCHRONOUS standby replica in a different AZ
-  All writes to primary are instantly replicated to standby
-  If primary fails → automatic DNS failover in ~1-2 minutes
-  Application connection string stays the SAME (DNS-based)
-  Standby is NOT accessible for reads — it is DR-only
-  Protects against: AZ failure, hardware failure, maintenance

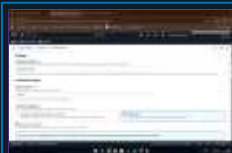
Multi-AZ vs Read Replica

Feature	Multi-AZ	Read Replica
Purpose	HA/DR	Performance
Sync	Sync	Async
Readable?	No	Yes
Failover	Auto	Manual

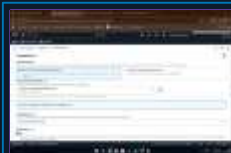
RDS Implementation Steps



MySQL Engine
Selected



DB Credentials
Configured



VPC & Subnet
Connectivity



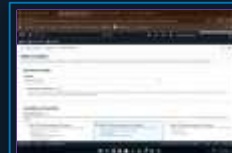
DB Status:
Available



Multi-AZ
Enabled



Snapshot
Available



Restore
Snapshot Page

Common Mistakes & Real Doubts Answered



EBS snapshot must be taken separately from AMI



AMI automatically creates EBS snapshots. You don't need to take them manually.



Public DB in RDS means anyone on the internet can access



Public = assigns a public IP. But Security Groups still control WHO can connect. Always restrict SGs.



Internet Gateway is assigned per subnet



IGW is attached to the VPC. Route tables at the subnet level control which subnets use it.



Snapshot stores the entire volume every time



Snapshots are INCREMENTAL. Only changed blocks are stored after the first full snapshot. Cost-efficient!



EBS volume can attach to EC2 in a different AZ



EBS is AZ-level storage. Volume and EC2 MUST be in the same AZ — no exceptions.



RDS works in any VPC without extra setup



RDS needs a DB Subnet Group with 2+ subnets in different AZs, plus DNS hostnames enabled in VPC.



Storing app data on the EC2 root disk is fine



Root disk is deleted when EC2 terminates (default). Always store app data on a separate EBS volume.



Multi-AZ standby can serve read traffic



Multi-AZ standby is for failover ONLY. It is not readable. Use Read Replicas for read scaling.

Interview Questions & Answers — Part 1

What is the difference between RTO and RPO?

RTO (Recovery Time Objective) defines how quickly the system must recover — it's about TIME. RPO (Recovery Point Objective) defines how much data loss is acceptable — it's about DATA. Example: RTO=15min means app must be back in 15 minutes; RPO=5min means we can lose up to 5 minutes of data.

What does an AMI contain? What does it NOT contain?

An AMI contains: OS, installed software, EC2 configuration, and EBS volume snapshots. It does NOT contain: VPC configuration, Security Groups, Elastic IPs, or IAM roles. These must be re-configured when launching a new EC2 from the AMI.

Explain RDS Multi-AZ. How is it different from Read Replicas?

Multi-AZ creates a synchronous standby replica in a different AZ for automatic failover (HA/DR). Read Replicas are asynchronous copies used for read scaling (performance). Multi-AZ standby cannot serve reads — it's purely for failover.

Where are EBS snapshots stored? Can you access the S3 bucket?

EBS snapshots are stored internally in Amazon S3, but it's managed by AWS. You cannot directly access the S3 bucket. You interact with snapshots only through the EC2 console or API. They are region-scoped but can be copied to other regions.

Can you attach an EBS volume to an EC2 in a different AZ?

No. EBS volumes are AZ-specific storage. You can only attach an EBS volume to an EC2 instance in the SAME Availability Zone. To move data across AZs, take a snapshot and create a new volume in the target AZ from that snapshot.

Interview Questions & Answers — Part 2

Why does RDS require multiple subnets?

RDS requires a DB Subnet Group with at least 2 subnets in different Availability Zones. This is mandatory even for Single-AZ deployments — AWS needs multiple AZs available to provision and manage the DB instance reliably.

What are the 4 DR strategies in AWS? When do you use each?

1) Backup & Restore — cheapest, hours of recovery, for non-critical apps. 2) Pilot Light — minimal DB running in DR, minutes-hours recovery. 3) Warm Standby — small full env running, minutes recovery for business-critical. 4) Active-Active — full infra both regions, seconds recovery for mission-critical.

What is the difference between Backup, High Availability, and Fault Tolerance?

Backup = copy of data stored safely (you still need to restore it — there IS downtime). High Availability = system continues with minimal interruption if one component fails (automatic failover). Fault Tolerance = zero downtime even during failure (full redundancy).

If an EC2 is terminated, is the data on its attached EBS volume lost?

It depends on the volume type. The ROOT volume (default) is deleted on termination. Additional data volumes that were attached AFTER instance launch persist by default. You can also modify root volume settings to persist — uncheck 'Delete on Termination' when launching.

Why would you use Route53 in a DR architecture?

Route53 is AWS's DNS service. In a DR setup, you configure health checks on your primary endpoint. If health check fails, Route53 automatically routes traffic to the DR region endpoint. This DNS-based failover is how multi-region architectures achieve near-zero downtime during regional failures.

Services Used & Why — The Architecture Decisions



Amazon EC2

WHY: Compute layer for production workloads. EC2 can be backed up as AMI and recovered in minutes.

PROD: Run web servers, app servers, microservices. Auto Scaling adds DR capability.



Amazon EBS

WHY: Block storage that acts as a virtual hard disk. Needed because apps require a filesystem, not object storage like S3.

PROD: Databases (MySQL on EC2), log storage, application data that needs IOPS performance.



EBS Snapshots

WHY: Incremental point-in-time backups of EBS volumes stored durably in S3. Enables recovery from any point.

PROD: Daily automated snapshots via AWS Backup or Data Lifecycle Manager. Retained for 30-90 days.



AMI

WHY: Full server image enables rapid EC2 recovery without manual configuration. Contains OS + apps + EBS snapshots.

PROD: Golden AMI strategy: create base AMI, bake apps in, use for all deployments. Cross-region copy for DR.



Amazon RDS

WHY: Managed database eliminates DBA overhead. Handles patches, backups, Multi-AZ replication automatically.

PROD: Production OLTP databases. Auto backup + point-in-time restore within backup retention period (1-35 days).



RDS Multi-AZ

WHY: Automatic database failover in ~1-2 minutes. Application connection string unchanged during failover.

PROD: Mandatory for any production database. RPO ~0 (synchronous replication). RTO ~1-2 minutes.

Production Best Practices

Backup Strategy

- ✓ Automate EBS snapshots via AWS Data Lifecycle Manager
- ✓ Enable automated RDS backups (retention: 7-35 days)
- ✓ Create AMIs after every major deployment — keep versioned
- ✓ Copy snapshots and AMIs to a DR region
- ✓ Test restores quarterly — backups that aren't tested aren't real

High Availability

- ✓ Always use RDS Multi-AZ for production databases
- ✓ Deploy EC2 behind Application Load Balancer across multiple AZs
- ✓ Use Auto Scaling Groups for automatic EC2 replacement
- ✓ Enable deletion protection on RDS and critical EBS volumes
- ✓ Use Route53 health checks for DNS-based failover

Security & Compliance

- ✓ Enable EBS encryption at rest (KMS) for all volumes
- ✓ Enable RDS encryption for databases handling PII/sensitive data
- ✓ Never make RDS publicly accessible in production — use VPN/Bastion
- ✓ Use IAM roles (not users) for EC2-to-AWS service access
- ✓ Enable CloudTrail + CloudWatch for audit and monitoring

 **Production Reality:** The largest cost of downtime is not the AWS bill — it's the revenue loss, SLA penalties, and customer trust erosion. A DR investment of \$500/month can prevent a \$300,000/hour outage.

Future Improvements — Production-Ready Enhancements

Key Learning Outcomes



3-Layer DR Architecture

Can design and implement Compute (AMI), Storage (EBS Snapshots), and Database (RDS Multi-AZ + Snapshots) recovery independently and as a unified DR strategy.



RTO & RPO Planning

Understand how to map business requirements to technical DR choices. Know which AWS strategy (Backup/Restore vs Active-Active) matches which RTO/RPO target.



RDS High Availability

Configured Multi-AZ deployment, understood synchronous replication, automatic DNS failover, and the real-world difference between Multi-AZ and Read Replicas.



EBS Lifecycle Management

Hands-on: Created volume, formatted, mounted, wrote data, took snapshot, restored volume, verified data. Full backup-restore lifecycle proven in production simulation.



DR Region Strategy

Understand that complete DR requires replicating compute, storage, and database layers to a second AWS region. AMI copies + snapshot copies + RDS cross-region read replicas.



Interview Readiness

Can answer 20+ common cloud interview questions on DR, HA, fault tolerance, EBS, AMI, RDS, snapshots with real hands-on evidence from this project.

Project Complete

Architecting Disaster Recovery & High Availability on AWS

- ✓ EBS Volume lifecycle — create, attach, format, mount, write, snapshot, restore
- ✓ EC2 AMI backup — created AMI, simulated disaster, recovered from AMI
- ✓ RDS MySQL — deployed with full configuration, connectivity, VPC setup
- ✓ RDS Multi-AZ — enabled synchronous standby, understood automatic failover
- ✓ RDS Snapshot — manual snapshot created, restore workflow demonstrated
- ✓ DR Strategies — Backup & Restore, Pilot Light, Warm Standby, Active-Active explained

Repository Structure

```
├─ README.md
├─ architecture/
│   └─ arc.png
├─ implementation-guide/
├─ screenshots/
├─ video-demo/
├─ notes/
└─ documentation/
```

*Built with native AWS services only. No third-party tools.
Production-grade · Portfolio-ready · Interview-focused*